



Date: June 4, 2025

Trimester 3
Practice Test 3

Name:

Class of:

Teacher:

Homerroom:

1. Which of the following functions exhibit exponential decay? Select **all** that apply.

$$y = ab^x \Rightarrow EF$$

only multiplies/ base matters

if $b > 1 \Rightarrow$ growth

if $b < 1 \Rightarrow$ decay

growth

decay

decay

growth

~~A.~~ $g(x) = 5 \left(\frac{4}{3}\right)^x$

B. $g(x) = \frac{3}{2} \left(\frac{3}{5}\right)^x$

C. $g(x) = -7 \left(\frac{1}{3}\right)^{2x}$

~~D.~~ $g(x) = -\frac{2}{3} (4)^x$

2. A lake now contains 800 trout and 300 bass. The trout population is decreasing at an annual rate of 4%, while the bass population is increasing at a continuous rate of 3% per year. After how many years will the number of bass first exceed the number of trout?

Bass $\left\{ \begin{aligned} A &= P \left(1 + \frac{r}{n}\right)^{nt} \\ A &= 800 \left(1 + \frac{-0.04}{1}\right)^{1 \cdot t} \\ A &= 800(0.96)^t \end{aligned} \right.$

Bass $\left\{ \begin{aligned} A &= P e^{rt} \\ A &= 300 e^{0.03t} \end{aligned} \right.$

~~A.~~ 11.1 years

B. 13.9 years

~~C.~~ 15.6 years

~~D.~~ 17.9 years

Plug into calculator

$Y1: 800(.96)^x$

$Y2: 300(e^{0.03x})$

Check the table

By $t=14$, Bass > Trout $\Rightarrow$ So the answer is 13.9 [B]

3. Which of the following statements is true?

~~A.~~ $\log_b m + \log_b n = \log_b(m+n)$ | $\log_b m + \log_b n = \log_b [m \cdot n] = \log_b mn \neq \log_b [m+n]$

B. $-2 \log_k p = \log_k \frac{1}{p^2}$ | $-2 \log_k p = \log_k p^{-2} = \log_k \frac{1}{p^2}$

~~C.~~ $\log_a(xy) = \frac{\log_a x}{\log_a y}$ | $\log_a(xy) = \frac{\log_a x}{\log_a a} \neq \frac{\log_a x}{\log_a y}$

~~D.~~ $\log_c 7 = \frac{\log 7}{\log c} - 1$ | $\log_c 7 = \frac{\log 7}{\log c}$

4. Jaden graphed the function $h(x) = \log_2 x$ and wants to decrease its rate of change. Which of the following graphs achieves that goal?

$$y = a \cdot b^x$$

$\left\{ \begin{aligned} \text{if } a < 1 &\Rightarrow \text{decreasing rate} \\ b \text{ is larger} &\Rightarrow \text{decreasing rate} \end{aligned} \right.$

~~A.~~ $h(x) = \log_2\left(\frac{1}{2}x\right)$

B. $h(x) = \frac{1}{8} \log_2 x$ $\rightarrow$ initial value is lower $\Rightarrow$ decreasing rate

~~C.~~ $h(x) = 4 \log_2 x$

~~D.~~ $h(x) = \log_4 x$ $\rightarrow$ higher base $\rightarrow$ decreasing rate

5. If $(ax+3)(bx+5) = 20x^2 + kx + 15$ for all values of x , and $a+b=9$, what are the two possible values for k ?

Step 1

$$\begin{array}{|c|c|c|} \hline ax & 3 \\ \hline abx^2 & 5bx \\ \hline 5ax & 15 \\ \hline \end{array}$$

$$\Rightarrow abx^2 + 5ax + 3bx + 15 = 20x^2 + kx + 15$$

$$ab = 20 \quad 5a + 3b = kx$$

$$15 = 15$$

~~A.~~ 12 and 28

~~B.~~ 24 and 30

C. 35 and 37

~~D.~~ 40 and 50

Step 2

$$ab = 20$$

$$a + b = 9$$

$$4 \cdot 5 = 20$$

$$4 + 5 = 9$$

$$\text{So, } a = 4, b = 5$$

$$5ax + 3bx = kx$$

$$5(4)x + 3(5)x = kx$$

$$\Rightarrow 20x + 15x = kx$$

$$\Rightarrow 35x = kx$$

$$k = 35$$

6. Which statement is not always true?

IR/IR = don't know $\rightarrow$ A. The quotient of two irrational numbers is irrational. $\Rightarrow$ No

~~B.~~ The product of two integers is an integer. $\Rightarrow$ Yes

~~C.~~ The sum of two fractions is a fraction. $\Rightarrow$ Yes

~~D.~~ The sum of a rational and an irrational number is irrational. $\Rightarrow$ Yes

$$\Rightarrow \frac{\sqrt{3}}{\sqrt{3}} = 1 \rightarrow R$$

7. Vector $\vec{u}$ has magnitude 15 and makes a 30° angle with the horizontal. Vector $\vec{v}$ has magnitude 6 and makes a 70° angle with the horizontal. What is the magnitude of $\vec{u} + \vec{v}$?

$$\begin{aligned} u_x &= 15 \cos 30^\circ & R_x &= u_x + v_x = 15.042 \\ u_y &= 15 \sin 30^\circ & R_y &= u_y + v_y = 13.138 \\ v_x &= 6 \cos 70^\circ & R &= \sqrt{R_x^2 + R_y^2} = 19.97 \approx 20 \\ v_y &= 6 \sin 70^\circ & & \end{aligned}$$

~~A.~~ 19.4 units
☒ B. 20.0 units
~~C.~~ 20.6 units
~~D.~~ 21.3 units

8. Find s to the nearest thousandth if $7^{-10s+8} = 18$.

$$\log_7(7^{-10s+8}) = \log_7 18$$

$$\Rightarrow -10s + 8 = \log_7 18 \Rightarrow s = \frac{\log_7 18 - 8}{-10}$$

~~A.~~ $s = 0.383$
~~B.~~ $s = 0.421$
~~C.~~ $s = 0.438$
☒ D. $s = 0.652$

9. Which of the following statements is true?

- ~~A.~~ The axis of symmetry of $y = ax^2 + bx + c$ is $x = -\frac{b}{a}$. No, it's $x = \frac{-b}{2a}$
~~B.~~ A quadratic with negative leading coefficient opens upward. → opens downward
☒ C. A quadratic can have zero, one, or two x -intercepts. → true
~~D.~~ The y -intercept of a quadratic is always positive. →  negative intercept

10. Which of the following functions represent exponential growth? Select **all** that apply.

- ☒ A. $g(x) = \left(\frac{5}{4}\right)^x$ $\frac{5}{4} > 1 \Rightarrow$ growth
~~B.~~ $g(x) = 2\left(\frac{2}{3}\right)^x$ $\frac{2}{3} < 1 \Rightarrow$ decay
☒ C. $g(x) = -\frac{3}{4}\left(\frac{3}{2}\right)^x$ $\frac{3}{2} > 1 \Rightarrow$ growth
~~D.~~ $g(x) = 7(0.9)^x$ $0.9 < 1 \Rightarrow$ decay

1. Which of the following functions exhibit exponential decay? Select **all** that apply.

- A. $g(x) = 5 \left(\frac{4}{3}\right)^x$
 B. $g(x) = \frac{3}{2} \left(\frac{3}{5}\right)^x$
 C. $g(x) = -7 \left(\frac{1}{3}\right)^{2x}$
 D. $g(x) = -\frac{2}{3} (4)^x$

Solution. Exponential decay requires a base satisfying $0 < b < 1$.

- A: base $\frac{4}{3} > 1 \Rightarrow$ growth.
- B: base $\frac{3}{5} < 1 \Rightarrow$ decay. ✓
- C: $(\frac{1}{3})^{2x} = (\frac{1}{9})^x$ with base < 1 (overall sign is irrelevant for decay of magnitude). ✓
- D: base $4 > 1 \Rightarrow$ growth.

B, C

2. A lake has 800 trout decreasing 4% per year and 300 bass growing continuously at 3% per year. After how many years will bass outnumber trout?

- A. 11.1 years
 B. 13.9 years
 C. 15.6 years
 D. 17.9 years

Solution.

$$800(0.96)^t = 300e^{0.03t} \implies e^{0.03t}(0.96)^{-t} = \frac{8}{3} \implies e^{0.070821t} = 2.6667 \implies t = \frac{\ln(2.6667)}{0.070821} \approx 13.9 \text{ years.}$$

B

3. Which of the following statements is true?

- A. $\log_b m + \log_b n = \log_b(m + n)$
 B. $-2 \log_k p = \log_k \frac{1}{p^2}$
 C. $\log_a(xy) = \frac{\log_a x}{\log_a y}$
 D. $\log_c 7 = \frac{\log 7}{\log c} - 1$

Solution. Only statement B correctly applies the power rule: $\log_k(p^{-2}) = -2 \log_k p$. The other three misuse logarithm laws.

B

4. Jaden graphed $h(x) = \log_2 x$ and wants to decrease its rate of change. Which graph accomplishes this?

- A. $h(x) = \log_2(\frac{1}{2}x)$
 B. $h(x) = \frac{1}{8} \log_2 x$
 C. $h(x) = 4 \log_2 x$
 D. $h(x) = \log_4 x$

Solution. Multiplying by a factor $0 < k < 1$ (B) or choosing a larger base (D) both flatten the graph, reducing its slope. Choice A merely translates the graph; choice C makes it steeper.

B and D

5. If $(ax + 3)(bx + 5) = 20x^2 + kx + 15$ for all values of x , and $a + b = 9$, what are the two possible values for k ?

- A. 12 and 28
- B. 24 and 30
- C. 35 and 37
- D. 40 and 50

Solution.

Step 1: Match the leading coefficient. Expanding $(ax + 3)(bx + 5)$ gives

$$abx^2 + (5a + 3b)x + 15.$$

Because the coefficient of x^2 on the right-hand side is 20,

$$ab = 20.$$

Step 2: Use the sum condition. The problem states $a + b = 9$. We need two real numbers whose product is 20 and sum is 9:

$$\{a, b\} = \{4, 5\}.$$

Step 3: Find the coefficient k . For $(a, b) = (4, 5)$:

$$k = 5a + 3b = 5(4) + 3(5) = 20 + 15 = 35.$$

For $(a, b) = (5, 4)$:

$$k = 5a + 3b = 5(5) + 3(4) = 25 + 12 = 37.$$

C

6. Which statement is not always true?

- A. The quotient of two irrational numbers is irrational.
- B. The product of two integers is an integer.
- C. The sum of two fractions is a fraction.
- D. The sum of a rational and an irrational number is irrational.

Solution. $\sqrt{2}/\sqrt{2} = 1$ is a rational quotient of two irrationals, so A fails. Statements B, C, D are always true.

A

7. $|\vec{u}| = 15$, $\theta_u = 30^\circ$; $|\vec{v}| = 6$, $\theta_v = 70^\circ$. Find $|\vec{u} + \vec{v}|$.

- A. 19.4
- B. 20.0
- C. 20.6
- D. 21.3

Solution. Component decomposition.

$$\vec{u} = \langle 15 \cos 30^\circ, 15 \sin 30^\circ \rangle = \langle 15(\frac{\sqrt{3}}{2}), 15(\frac{1}{2}) \rangle = \langle 12.990, 7.500 \rangle,$$

$$\vec{v} = \langle 6 \cos 70^\circ, 6 \sin 70^\circ \rangle = \langle 6(0.3420), 6(0.9397) \rangle = \langle 2.052, 5.638 \rangle.$$

Add the components.

$$\vec{u} + \vec{v} = \langle 12.990 + 2.052, 7.500 + 5.638 \rangle = \langle 15.042, 13.138 \rangle.$$

Magnitude of the resultant.

$$|\vec{u} + \vec{v}| = \sqrt{(15.042)^2 + (13.138)^2} = \sqrt{226.3 + 172.6} = \sqrt{398.9} \approx 19.97 \text{ units.}$$

19.97 rounds to 20.0 units, which matches option **B**.

8. Solve $7^{-10s+8} = 18$ to the nearest thousandth.

- A. 0.383 B. 0.421
C. 0.438 D. 0.652

Solution.

$$-10s + 8 = \frac{\ln 18}{\ln 7} \approx 1.486 \implies s = \frac{8 - 1.486}{10} \approx 0.652.$$

D

~~No listed choice matches 0.652. None of the answers is correct.~~

9. Which statement is true?

- A. $x = -\frac{b}{a}$ is the axis of symmetry of $y = ax^2 + bx + c$
B. A quadratic with negative leading coefficient opens upward.
C. A quadratic can have zero, one, or two x -intercepts.
D. The y -intercept of a quadratic is always positive.

Solution. Option C is correct; the others are false ($x = -b/2a$, downward opening for negative a , and the y -intercept is c , which may be any real number).

C

10. Which of the following functions represent exponential growth? Select **all** that apply.

- A. $g(x) = \left(\frac{5}{4}\right)^x$
B. $g(x) = 2\left(\frac{2}{3}\right)^x$
C. $g(x) = -\frac{3}{4}\left(\frac{3}{2}\right)^x$
D. $g(x) = 7(0.9)^x$

Solution. Growth requires $|\text{base}| > 1$.

- A: base $1.25 > 1 \Rightarrow$ growth. ✓
- B: base $0.667 < 1$ decay.
- C: base $1.5 > 1$ (overall sign negative but magnitude grows). ✓
- D: base $0.9 < 1$ decay.

A, C